Comprehensive AI-Powered Dental Diagnostic & Treatment System

Problem Statement Document

1. Background

Dental healthcare is one of the most frequently required medical services globally, yet accurate and timely diagnosis remains a significant challenge in many clinical settings. Dentists must assess multiple oral conditions simultaneously — including tooth decay (caries), misaligned teeth (malocclusion), oral wounds, and calculus (tartar buildup) — using manual visual examination. This approach is inherently limited by human perception, time pressure, and the absence of a unified diagnostic tool.

In Sri Lanka, most dental clinics rely entirely on naked-eye inspection with no digital assistance. Each condition typically requires separate examination methods, separate documentation, and separate clinical decisions. This fragmented process is time-consuming, inconsistent across practitioners, and prone to missed early-stage diagnoses.

This project addresses that gap by building a comprehensive AI-powered dental diagnostic and treatment system that integrates four simultaneous detection capabilities into a single unified platform — enabling dentists to assess a patient's complete oral health from a single photograph or X-ray input.

2. Problem Being Solved

Dental clinics that operate without AI-assisted diagnostic tools face several recurring problems that affect both the quality of patient care and the efficiency of clinical workflows.

2.1 Single-Condition Diagnostic Tools Create Inefficiency

Current AI or digital diagnostic tools in dentistry are designed to detect only one condition at a time. A dentist who wants to check for both caries and calculus must use two separate tools, upload the same image twice, and manually reconcile two separate outputs. This workflow is slow, repetitive, and error-prone. Our system processes a single dental photograph or X-ray through a unified Multi-Condition Dental Diagnostic Engine (MC-DDE) that detects all four conditions simultaneously in under three seconds.

2.2 Early-Stage Conditions Are Missed by Naked-Eye Examination

Tooth decay in its early stages, minor calculus deposits, and small oral wounds are frequently invisible to the naked eye, especially under standard clinic lighting. Delayed detection leads to more invasive and expensive treatment. Our deep learning models are trained on thousands of labeled clinical images to detect sub-visual patterns that indicate early disease, improving diagnostic sensitivity beyond what manual inspection can achieve.

2.3 No Structured Patient Medical Record Management

Patient records in many clinics are stored in paper files or basic spreadsheets. There is no structured digital system to link diagnostic images to patient histories, track how conditions evolve over time, or retrieve past records quickly. Our system includes a dedicated Medical Resource and History Vault that stores all records — including X-rays, photos, reports, schedules, and inventory — in a structured MongoDB database with full CRUD management and soft-delete audit trails.

2.4 Treatment Planning Is Manual and Inconsistent

After diagnosis, dentists manually decide on treatment plans, estimate costs, and communicate options to patients verbally. There is no standard, data-driven process. This leads to inconsistencies between practitioners and reduces patient confidence. Our system's Predictive Analytics module uses detection results to recommend specific procedure codes, simulate disease progression over six months if untreated, and generate automated clinical reports with treatment recommendations.

2.5 Patient Communication and Engagement Is Poor

Patients often leave a dental appointment without clearly understanding their diagnosis, what their condition looks like, or why treatment is necessary. This reduces treatment acceptance rates. Our system generates patient-friendly visual outputs — including colour overlays showing caries depth, heatmaps for calculus density, and wound boundary highlights — alongside a simulated after-treatment view so patients can see the expected outcome before committing to a procedure.

2.6 No Integrated Appointment, Prescription, and Feedback Loop

Dental clinics manage appointments, prescriptions, medication orders, and patient feedback through disconnected manual systems. Staff must cross-reference multiple sources to get a complete picture of a patient's status. Our system integrates appointment scheduling, digital prescriptions, medication ordering with e-payment, and a feedback and rating module into one cohesive platform, with all data linked through a shared patient identifier.

3. Who This System Is For

User Type	What They Can Do
Patient / Guest	• Register an account, browse available doctors, check appointment availability • Book appointments and view full booking history • Receive AI-assisted diagnostic results and clinical reports • View and manage personal medical record history • Submit feedback and ratings for completed consultations • Manage emergency contact details
Dentist / Doctor	• Create and manage digital prescriptions with medications and dosages • View patient medical records and AI diagnostic results • Update appointment schedules and availability slots • Access AI-generated treatment recommendations and procedure codes
Administrator	• Manage all system records — rooms, facilities, and users • Verify medication orders and process payments • Process refunds and manage support tickets • Authorise deletion of outdated medical records • Oversee all bookings and system-wide operations
AI/ML Engine (MC-DDE)	• Accept dental image inputs — photographs and X-rays • Run simultaneous four-condition inference pipeline • Return structured diagnostic results with severity classifications • Generate visual overlays and unified dental chart output

4. What the System Does

User Authentication and Profile Management

Users can register and log in securely. Each user maintains a profile with personal details and a profile image. Patients can view their booked consultations and update their personal information. Accounts can be deactivated or deleted by the user or an administrator.

AI-Powered Multi-Condition Diagnosis (MC-DDE)

The core AI feature accepts a single dental photograph or X-ray and simultaneously detects four conditions: caries, malocclusion, oral wounds, and calculus. Each condition is classified by severity and returned in under three seconds. Results are displayed through interactive visual overlays on a unified dental chart showing all four conditions at once.

Medical Resource and History Vault

All patient records are stored and managed through a structured CRUD module. Record types include X-rays, photos, clinical reports, doctor schedules, and inventory entries. Records are linked to the AI engine via a file URL field that enables image inference. Soft deletion ensures audit trails are preserved and admin authorisation is required for permanent removal.

Doctor's Digital Prescription Module

Dentists create structured digital prescriptions with selected medications and dosages. Prescriptions can be modified before final submission and deleted while in draft state. Patients can view their full prescription history through the patient portal.

Appointment Scheduling

Patients book appointments by selecting an available doctor, date, and time slot. The system prevents double bookings by checking for slot conflicts before confirming. Upcoming appointments can be rescheduled within the allowed timeframe and deleted when necessary.

Medication Ordering and E-Payment Gateway

Patients order medications directly from approved prescriptions. The system records card payment details, validates the format, and tracks order status from pending through to confirmed after admin verification. Unpaid orders can be cancelled by the patient at any time.

Feedback and Rating

After a completed consultation, patients can submit a star rating and written feedback comment. Feedback can be edited within an allowed time window and deleted thereafter. Doctors can view their aggregate rating summary alongside individual feedback entries.

Emergency Contact Management

Patients can register emergency contacts with name, relationship, and phone number. Contacts can be prioritised, updated, or removed at any time through the patient profile interface.

Predictive Analytics and Clinical Reporting

The system uses AI detection data to simulate how untreated conditions will progress over six months, visualise expected treatment outcomes, recommend specific dental procedure codes, and generate automated clinical reports that can be shared with patients to improve treatment acceptance.

5. Key Technical Decisions

Decision	Reason
React Native + Expo	Builds a single codebase that runs on both Android and iOS. Expo simplifies development with built-in camera, image picker, and device testing tools.
Node.js — built-in http module	Lightweight backend with no Express dependency. Demonstrates direct control over request routing, CORS handling, and body parsing.
MongoDB Atlas	Cloud-hosted document database suited to flexible medical record structures. Easy to scale and accessible from any deployment environment.
Mongoose ODM	Enforces schema validation, provides soft-delete patterns, and simplifies query building on top of MongoDB.
JWT Authentication	Stateless tokens mean the server does not store session data. Secure and scalable for mobile applications.
Python — AI/ML Layer	Industry-standard for deep learning. Libraries including TensorFlow, PyTorch, and scikit-learn provide the full model development and evaluation pipeline.
CNN / ResNet50 / EfficientNet	Pre-trained architectures accelerate training on dental image datasets and provide strong baseline performance for classification and detection.
U-Net Segmentation	Specialised architecture for boundary detection — used for oral wound segmentation and calculus deposit density mapping.
Render Deployment	Free cloud hosting for the backend with automatic deployment from GitHub. Accessible from any device via a public URL.
No Real Payment Gateway	Integration with a live payment provider is out of scope for this academic phase. Card format validation simulates the user experience without real transactions.

6. System Workflow Summary

Diagnostic Workflow

1. Patient or dentist logs in to the mobile application.
2. A dental photograph or X-ray is uploaded through the app interface.
3. The image is sent to the MC-DDE API endpoint.
4. The engine runs four parallel inference pipelines: caries detection, malocclusion classification, oral wound segmentation, and calculus mapping.
5. Results are returned with severity classifications and confidence scores for each condition.
6. The system generates visual overlays, a unified dental chart, and a clinical report with treatment recommendations.
7. The record is saved to the Medical Resource and History Vault, linked to the patient's ID.

8. The patient receives a patient-friendly visualisation and, if untreated, a six-month disease progression simulation.

Booking and Prescription Workflow

9. Patient logs in and browses available doctors and appointment slots.
10. Patient selects a date and time slot. The system checks for conflicts before confirming.
11. After the consultation, the dentist creates a digital prescription.
12. Patient views the prescription and places a medication order.
13. Payment details are submitted and validated for correct format.
14. Order status is updated from pending through to confirmed after admin verification.
15. Patient submits a feedback rating for the completed consultation.

Refund and Support Workflow

16. Patient submits a support ticket linked to a specific booking or order.
17. Administrator reviews the ticket and checks applicable policy rules.
18. If the policy allows it, the administrator processes the refund through a single verified action.
19. All related records are updated simultaneously: booking cancelled, payment refunded, ticket closed.

7. Scope and Limitations

Within Scope

- User registration, login, and full profile management
- AI/ML engine with four simultaneous dental condition detection models (MC-DDE)
- Medical record CRUD with soft-delete and admin authorisation
- Digital prescription creation, viewing, and management
- Real-time appointment scheduling with conflict prevention
- Medication ordering with mock payment submission and admin verification
- Feedback and rating system for completed consultations
- Emergency contact management
- Predictive analytics and automated clinical report generation
- Patient-friendly diagnostic visualisations and treatment outcome simulation
- Image upload for facility and diagnostic records
- Hosted backend accessible via public URL

Outside Scope — Current Academic Phase

- Real payment gateway integration (e.g., Stripe, PayHere)
- Real SMS or push notification provider — notifications are simulated
- Integration with physical dental equipment or DICOM imaging systems
- Multi-language support
- Automated recurring appointment subscriptions

- Regulatory compliance certification (e.g., HIPAA, HL7 FHIR)